Alkane

81. Which of the following fractions of petroleum refining contains kerosene ? (Boiling ranges in ^{o}C are given below)

(a) 40 - 80 (b) 80 - 200

(c) 200 - 300 (d) Above 300

82. Which of the following statements is incorrect ? The members of the homologous series of alkanes

(a) Are all straight chain compounds

(b)Have the general formula C_nH_{2n+2}

(c)Have similar chemical properties

(d) Show a regular gradation of physical properties

83. On mixing tetraethyl lead to gasoline available at petrol pumps

(a) Calorific value of the fuel I ncreases

(b) Odour diminishes

(c) Less smoke is obtained on combustion

(d) Antiknock property of fuel increases

84. A liquid hydrocarbon can be converted to gaseous hydrocarbon by

(a) Cracking

(b) Hydrolysis

(c) Oxidation

(d) Distillation under reduced pressure

85. The tetrahedral nature of carbon was first given by

(a) Kekule

(b) Le Bell and Van't Hoff

(c) Pauling

(d) Armstrong and Bayer

86. Formation of alkane by the action of *Zn* on alkyl halide is called

(a) Frankland's reaction

(b) Wurtz reaction

(c) Cannizzaro reaction

(d) Kolbe's reaction

87. Which of the following compounds will form a hydrocarbon on reaction with Grignard reagent

(a) CH_3CH_2OH

(b) CH_3CHO

(c) CH_3COCH_3

(d) $CH_3CO_2CH_2$

88. Name the hydrocarbon that is a liquid at STP

(a) Ethane (b) Propane
(c) n-butane (d) *n*-pentane

89. Which statement is not true concerning alkanes
(a) Large number alkanes are soluble in water
(b) All alkanes have a lower density than water
(c) At room temperature some alkanes are liquids, some solids and some gases
(d) All alkanes burn

90. Fischer Tropsch process is used for the manufacture of
(a) Synthetic petrol
(b) Thermosetting plastics
(c) Ethanol
(d) Benzene

91. Which one of the following compounds cannot be prepared by Wurtz reaction
(a) CH_4 (b) C_2H_6
(c) C_3H_8 (d) C_4H_{10}

92. A fuel contains 25 % *n*-heptane and 75 % iso-octane. Its octane number is
(a) 50 (b) 75
(c) 100 (d) 25

93. Sodium ethoxide is a specific reagent for
(a) Dehydration
(b) Dehydrogenation
(c) Dehydrohalogenation
(d) Dehalogenation

94. Which of the following has highest percentage of hydrogen
(a) CH_4 (b) C_2H_4
(c) C_6H_6 (d) C_2H_2

95. What is the molecular formula of the alkane, the 5.6 *litre* of which weight 11 *g* at STP
(a) C_6H_{14} (b) C_4H_{10}
(c) C_3H_8 (d) C_2H_6

96. The reference compound `iso-octane' which is used in determining the octane number of gasoline has the structure
(a) $CH_3-CH(CH_3)-CH(CH_3)-CH(CH_3)-CH_3$
(b) $CH_3-C(CH_3)_2-CH_2-CH(CH_3)-CH_3$
(c) $CH_3-C(CH_3)_2-CH(CH_3)-CH_2-CH_3$
(d) $CH_3-C(CH_3)_2-C(CH_3)_2-CH_3$

97. Sample of 2, 3-dibromo-3-methylpentane is heated with zinc dust. The resulting product is isolated and heated with *HI* in the presence of phosphorus. Indicate which is the structure that represent the final organic product formed in the reaction

(a) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$

(b) $CH_2=CH-\overset{\overset{CH_3}{|}}{CH}-CH_2-CH_3$

(c) $CH_3-CHI-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$

(d) $CH_2=CH-\underset{\underset{CH_3}{|}}{C(I)}-CH_2-CH_3$

98. The order of appearance of the following with rising temperature during the refining of crude oil is

(a) Kerosene oil, gasoline, diesel

(b) Diesel, gasoline, kerosene oil

(c) Gasoline, diesel, kerosene oil

(d) Gasoline, kerosene oil, diesel

99. When sodium propionate is heated with soda lime, the main product is

(a) Ethane (b) Methane

(c) Propane (d) Butane

100. Gasoline is a mixture of alkanes with the number of carbon atoms

(a) C_3-C_5 (b) C_5-C_6

(c) C_6-C_8 (d) C_7-C_9